

AUTONOMOUS MULTI-POINT WATER QUALITY SURVEY ROBOT

Low-Cost Autonomous Floating Rover for Water Quality Monitoring

CSE461: Introduction to Robotics | Lab Project Poster | Group 01 | Section 5 | Summer 2026
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01 ABSTRACT

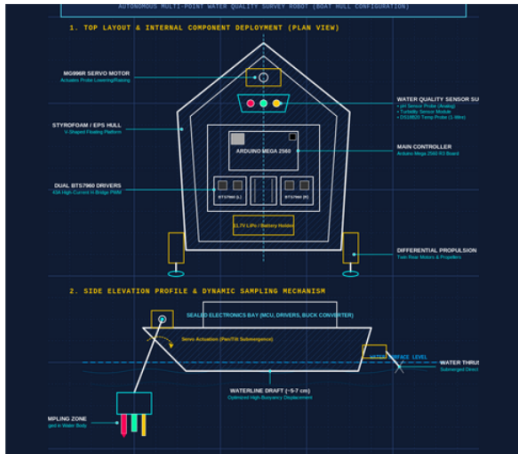
This project develops a low-cost autonomous floating rover for multi-point water-quality monitoring. Built around an Arduino Mega 2560, it measures pH, turbidity and temperature at predefined sampling points. Two DC geared motors with propellers provide propulsion, while an MG996R servo positions the sensor assembly. Readings are validated, shown on a 16x2 I2C LCD and stored on a microSD card for later analysis. The prototype demonstrates integrated sensing, control, navigation and data logging for environmental monitoring.

03 SYSTEM OVERVIEW

A. System Architecture — Draw.io PNG



B. Sketch / Concept Diagram — Floating Robot



C. Floating Structure

- Foam/float hull supports the electronics above water.
- Dual propellers provide water-surface propulsion.
- Sensor assembly is positioned by the MG996R servo.

02 INTRODUCTION

Background & Motivation

- Water pollution affects ecosystems, public health and aquatic resources.
- Manual multi-point sampling is labor-intensive and time-consuming.
- A floating rover can repeat measurements consistently.

Problem & Objectives

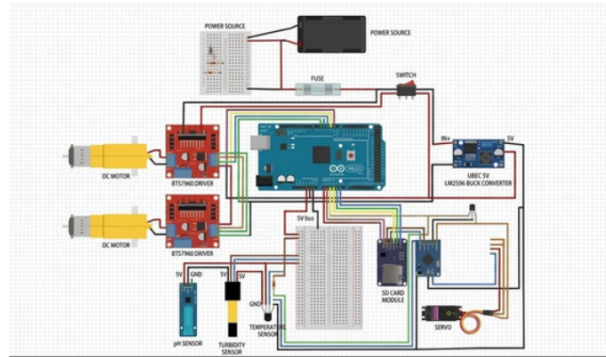
- Design a low-cost floating robot for multi-point surveys.
- Measure pH, turbidity and temperature at predefined points.
- Validate, display and store data automatically.
- Demonstrate movement between sampling points.

Real-World Relevance

- Applicable to ponds, lakes, reservoirs and similar water bodies.
- Supports environmental research and educational robotics.
- Provides an affordable platform for repeated local monitoring.

04 IMPLEMENTATION

A. Control Circuit Diagram — Digital Wiring Style



B. Hardware Components

Component	Function / Role
Arduino Mega 2560	Main controller
BTS7960 x2	Motor drivers
DC geared motors x2 + propellers	Propulsion
MG996R servo	Sensor positioning
pH sensor	pH measurement
Turbidity sensor	Turbidity measurement
DS18B20	Temperature measurement
16x2 I2C LCD	Local display
microSD module	Data logging
UBEC 5V + 11.7V battery	Power system

C. Actual Robot Prototype



05 FUNCTIONALITIES

Mechanical Functionality

- Floating platform with two propeller-driven DC geared motors.
- MG996R servo positions the sensor assembly.
- Electronics are mounted above the waterline.

Software Functionality

- Navigate → sample → read sensors → validate → store/display → next point.
- microSD logs readings; LCD shows local values.
- Predefined sampling sequence controls the mission.

Navigation Functionality

- Follows a predefined multi-point route without GPS.
- Movement uses programmed timing/sequence.
- Obstacle sensing is a future enhancement.

Software-Hardware Communication

- I2C for LCD.
- SPI for microSD.
- 1-Wire for DS18B20.
- Analog inputs for pH/turbidity; PWM for motors and servo.

06 APPLICATIONS & COMPETITORS

Applications

- Water-quality surveys in ponds, lakes and reservoirs.
- Environmental research and educational robotics.
- Repeated local monitoring and trend analysis.

Competitors / Alternatives

- Manual sampling: labor-intensive and limited by sampling frequency.
- Commercial autonomous platforms: generally higher cost and maintenance.
- Our prototype: low-cost, modular and suited to academic demonstration.

07 CONCLUSION

- Developed a low-cost autonomous floating rover for multi-point water-quality monitoring.
- Integrates sensing, propulsion, servo positioning, display and SD data logging.
- Demonstrates a modular robotics platform for environmental monitoring.
- Human supervision remains important for real-world deployment and safety.

KEY TAKEAWAY

Low-cost robotics can support repeatable water-quality surveys across multiple points.

08 FUTURE IMPROVEMENTS

- Add wireless telemetry (Wi-Fi / LoRa).
- Integrate GPS for precise geo-tagging and waypoint navigation.
- Add dissolved oxygen and other water-quality sensors.
- Improve battery capacity and explore solar charging.
- Add robust obstacle detection for field deployment.

09 LIMITATIONS

- Predefined route; no GPS in current prototype.
- Battery runtime limits mission duration.
- Sensor calibration is required periodically.

10 SAFETY CONSIDERATIONS

- Use a waterproof enclosure for electronics.
- Protect connectors and avoid short circuits.
- Handle the Li-ion battery safely.
- Operate only in suitable water conditions.

11 ACKNOWLEDGEMENT

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